

Estimating Urban Air Pollution Reduction Through Increased Use of Public Transport: A Data Science Approach

Introduction

Urban air pollution is a growing environmental and public health concern due to rapid urbanization and increasing transportation demands in cities. Rising population levels and daily commuting activities have significantly increased vehicle emissions, leading to poor air quality and health-related issues. Addressing this problem is essential for creating sustainable and healthy urban environments.

Problem Statement

One of the major contributors to urban air pollution is the excessive use of private vehicles for daily travel. High dependence on private transport increases traffic congestion and emission levels. This problem was chosen because transportation-related pollution directly affects urban air quality, and even small behavioural changes can lead to noticeable improvements.

Proposed Solution

This project uses a data-driven approach to analyse air quality indicators such as PM2.5, PM10, and AQI along with transport usage patterns. A predictive “what-if” analysis will be performed to estimate pollution reduction under different public transport adoption scenarios.

Conclusion / Expected Impact

The project aims to highlight how increased public transport usage can reduce urban air pollution and support sustainable transportation planning.

NAME – VINEET SINGH

ROLL NO – 25SCS1003005351

SECTION – 1CSE25

SIGNATURE _____